Jerry Meng

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Technical Skills

Languages: C#, TypeScript, Python, Java, C, C++, SQL, Bash
Frameworks: .NET, ASP.NET Core, EntityFramework Core, Springboot, React
Databases: PostgreSQL, SQL Server, MySQL, Redis
Cloud & DevOps: Azure, Azure Containers, Docker, GitHub Actions, Azure Pipelines, Azure Storage
Developer Tools: Git, SVN, Postman, Jira, Zoho

Experience

Software Developer Co-op | *C#, ASP.NET Core, MSSQL, MSTest, C++, Agile* May 2026 – Sept 2026
BC Hydro - Powertech Labs *Surrey, BC*

- Built REST APIs in an ASP.NET Core Web API for power flow diagram analysis
- Designed a Python document generation pipeline, adopted across 50+ engineers to automate technical and release documentation
- Validated MSSQL referential constraints by writing targeted EF Core queries, preventing improper data flows
- Refactored and improved code quality and scalability of a legacy MFC C++ desktop application for analysing short circuiting
- Wrote unit tests for REST controllers and data access layers within an Agile development environment

Fullstack Developer – Capstone | *Java, Springboot, Zoho, Spring JPA, UML* Jan 2026 – May 2026
Aether Automation, SFU *Surrey, BC*

- Co-developed a scalable Spring Boot microservice to automate architectural diagram generation, eliminating manual workflows for client project deliverables
- Engineered a secure data pipeline using Zoho Creator v2 REST APIs and OAuth 2.0 to reliably fetch and authenticate client CRM records
- Built a GenAI-powered XML generation engine, converting structured project data into import-ready Lucidchart diagrams
- Improved search engine visibility and SEO by using Server-Side Rendering (SSR) with Thymeleaf pages

Projects

Multi-User Collaborative Canvas | *C#, ASP.NET, SignalR, Azure Blob, PostgreSQL, React, Redis*

- Designed real-time synchronization for multi-user drawing and cursor interactions using SignalR
- Achieved low-latency broadcasting using SignalR web sockets, and utilizing a Redis backplane to scale messaging across multiple server instances
- Built ASP.NET Core REST APIs with Entity Framework Core for a React/Konva.js canvas client
- Eliminated database bottlenecks with background services that batch-insert drawing logs, using Redis to cache active session states
- Dockerized and deployed the application to Azure with Azure Pipelines for automated CI/CD and production releases

AI Rehabilitation Platform – Hackathon Track Winner | *TypeScript, React, MediaPipe, ElevenLabs, VAPI*

- Won multiple tracks at Mountain Madness 2026 by co-developing an AI-powered physical rehabilitation web application
- Integrated MediaPipe to capture and analyze live webcam coordinates, delivering real-time posture correction to users
- Orchestrated a voice guidance system using VAPI and ElevenLabs APIs, providing instant, spoken audio feedback based on the user's exercise form
- Leveraged Gemini API to process raw session movement data and generate PDF progress reports via jsPDF

Education

Simon Fraser University September 2024 – May 2028
Bachelor of Computing Science *Burnaby, BC*

- GPA: 3.57
- Courses: Systems Programming, Data Structures, Software Engineering Principles, Database Systems